Ultrafast Transient Grating Spectroscopy of GaAs

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Abstract

Transient-grating spectroscopy measures carrier relaxation and spatial diffusion through the decay of a photoinduced grating. The supplied instructions identify delay in picoseconds and specify that the sample grating period is one quarter of the phase-mask period in each filename. Biexponential fits to the averaged valid scans give fast and slow decay components; the slow finite-q rates yield a TG-only diffusion estimate of $(3.7 \pm 0.7) \times 10^{-3}\,\mathrm{m^2\,s^{-1}}$. The $q = 0$ pump–probe rate does not agree with the TG extrapolation, so this is an exploratory estimate. Selecting the lowest-normalized-residual complete autocorrelation trace gives pulse durations of 81.7 fs before and 79.0 fs after the compressor.

1 Theoretical background

An ultrashort pump pulse creates electron–hole pairs in GaAs. Two pump fields with equal intensity I_0 and wave vectors $\mathbf{k}_1, \mathbf{k}_2$ produce the intensity pattern

$$I(\mathbf{r}) = |E_1 + E_2|^2 \tag{1}$$
$$= |E_1|^2 + |E_2|^2 + E_1 E_2^* + E_1^* E_2 \tag{2}$$
$$= 2I_0 \left[1 + \cos\left((\mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r} + \phi\right)\right]. \tag{3}$$

For an angle θ between the beams, $q = |\mathbf{k}_1 - \mathbf{k}_2| = 2k \sin(\theta/2)$ and

$$\Lambda_{\mathrm{TG}} = \frac{2\pi}{q} = \frac{\lambda}{2\sin(\theta/2)}. \tag{4}$$

The two-beam interference and resulting spatial carrier modulation are illustrated in Fig. 1.

Write the carrier density as a uniform part plus its sinusoidal grating component,

$$n(x,t) = N_0(t) + N_q(t)\cos(qx). \tag{5}$$

Substitution into the tutorial's one-dimensional diffusion–recombination equation gives

$$\frac{\partial n(x,t)}{\partial t} = D_a \frac{\partial^2 n(x,t)}{\partial x^2} - \frac{n(x,t)}{\tau_{\mathrm{rec}}}. \tag{6}$$

Using $\partial_t n = \dot{N}_0 + \dot{N}_q \cos(qx)$ and $\partial_x^2 n = -q^2 N_q \cos(qx)$, and equating the coefficients of 1 and $\cos(qx)$, respectively,

$$\dot{N}_0 = -\frac{N_0}{\tau_{\mathrm{rec}}}, \qquad \dot{N}_q = -\left(D_a q^2 + \frac{1}{\tau_{\mathrm{rec}}}\right) N_q. \tag{7}$$

$$\int_{N_0(0)}^{N_0(t)} \frac{dN_0}{N_0} = -\int_0^t \frac{dt'}{\tau_{\mathrm{rec}}}, \qquad \int_{N_q(0)}^{N_q(t)} \frac{dN_q}{N_q} = -\int_0^t \left(D_a q^2 + \frac{1}{\tau_{\mathrm{rec}}}\right) dt'. \tag{8}$$

Evaluation yields

$$N_0(t) = N_0(0)e^{-t/\tau_{\rm rec}}, \tag{9}$$

$$N_q(t) = N_q(0)e^{-(D_a q^2 + 1/\tau_{\rm rec})t}. \tag{10}$$

For signals proportional to these populations, the $q = 0$ transient-reflectivity and finite-q grating traces therefore have the forms

$$S_{\rm pop}(t) = A_{\rm pop}e^{-t/\tau_{\rm rec}} + C_{\rm pop}, \tag{11}$$

$$S_{\rm TG}(t) = A_{\rm TG}e^{-t/\tau_{\rm TG}} + C_{\rm TG}, \tag{12}$$

$$\frac{1}{\tau_{\rm TG}} = \frac{1}{\tau_{\rm rec}} + D_a q^2. \tag{13}$$

In $1/\tau_{\rm TG} = 1/\tau_{\rm rec} + D_a q^2$, the intercept is $1/\tau_{\rm rec}$ and the slope is D_a.

The provided measurement instructions requests two decay components to represent faster carrier relaxation and slower processes such as grating diffusion. We therefore use the phenomenological extension

$$S(t, q) = C + A_f e^{-t/\tau_f} + A_s e^{-t/\tau_s}, \qquad 0 < \tau_f < \tau_s. \tag{14}$$

The assignments of the two fitted components are interpretations of their timescales. To estimate diffusion, the slow-component rate $1/\tau_s$ is regressed against q^2 for the finite-q TG measurements; the single-population diffusion equation above motivates this comparison, but does not by itself derive the biexponential response.

For ambipolar transport, set the excess electron and hole densities equal, $n = h$, and equate their particle fluxes. With the drift–diffusion expressions

$$\mathbf{j}_e = -D_e\nabla n - \mu_e n\mathbf{E}, \qquad \mathbf{j}_h = -D_h\nabla n + \mu_h n\mathbf{E}, \tag{15}$$

the condition $\mathbf{j}_e = \mathbf{j}_h$ gives

$$-(D_e - D_h)\nabla n = (\mu_e + \mu_h)n\mathbf{E}, \tag{16}$$

$$\mathbf{E} = \frac{D_h - D_e}{\mu_e + \mu_h}\frac{\nabla n}{n}, \tag{17}$$

$$\mathbf{j}_e = -\left[D_e + \mu_e\frac{D_h - D_e}{\mu_e + \mu_h}\right]\nabla n \tag{18}$$

$$= -\frac{D_e\mu_h + D_h\mu_e}{\mu_e + \mu_h}\nabla n. \tag{19}$$

Comparing with Fick's law $\mathbf{j} = -D_a\nabla n$ identifies

$$D_a = \frac{D_e\mu_h + D_h\mu_e}{\mu_e + \mu_h}, \tag{20}$$

$$D_i = \frac{k_B T}{e}\mu_i \quad (i = e, h) \tag{21}$$

$$\implies \quad D_a = \frac{2\mu_e\mu_h}{\mu_e + \mu_h}\frac{k_B T}{e} = \mu_a\frac{k_B T}{e}, \tag{22}$$

$$\mu_a = \frac{2\mu_e\mu_h}{\mu_e + \mu_h}. \tag{23}$$

The Einstein step assumes non-degenerate carriers at a common temperature. At high carrier density, many-body and degeneracy effects can change D_a [6].

For a Gaussian pulse intensity $I(t) = I_0 e^{-t^2/(2\sigma_t^2)}$, the intensity autocorrelation is

$$G(\tau) = \int_{-\infty}^{\infty} I(t) I(t-\tau)\, dt \tag{24}$$

$$= I_0^2 \int_{-\infty}^{\infty} \exp\left[-\frac{t^2 + (t-\tau)^2}{2\sigma_t^2}\right] dt \tag{25}$$

$$= I_0^2 e^{-\tau^2/(4\sigma_t^2)} \int_{-\infty}^{\infty} e^{-(t-\tau/2)^2/\sigma_t^2}\, dt \tag{26}$$

$$= I_0^2 \sqrt{\pi}\sigma_t e^{-\tau^2/(4\sigma_t^2)}. \tag{27}$$

Here $t^2 + (t-\tau)^2 = 2(t-\tau/2)^2 + \tau^2/2$ and the substitution $u = (t-\tau/2)/\sigma_t$ reduces the integral to $\sigma_t \int_{-\infty}^{\infty} e^{-u^2} du = \sqrt{\pi}\sigma_t$. The intensity half-maximum satisfies $e^{-t^2/(2\sigma_t^2)} = 1/2$, so $\Delta t_{\text{pulse}} = 2\sqrt{2\ln 2}\,\sigma_t$. The autocorrelation half-maximum satisfies $e^{-\tau^2/(4\sigma_t^2)} = 1/2$, so $\Delta t_{\text{AC}} = 4\sqrt{\ln 2}\,\sigma_t = \sqrt{2}\,\Delta t_{\text{pulse}}$.

The Gaussian time–bandwidth product follows by Fourier transforming the field envelope $a(t) = \sqrt{I_0} e^{-t^2/(4\sigma_t^2)}$:

$$\widetilde{a}(\nu) = \int_{-\infty}^{\infty} a(t) e^{-i2\pi\nu t}\, dt \tag{28}$$

$$= \sqrt{I_0}\, e^{-4\pi^2\sigma_t^2\nu^2} \int_{-\infty}^{\infty} \exp\left[-\frac{(t + i4\pi\sigma_t^2\nu)^2}{4\sigma_t^2}\right] dt \tag{29}$$

$$= 2\sqrt{\pi}\sigma_t \sqrt{I_0}\, e^{-4\pi^2\sigma_t^2\nu^2}, \tag{30}$$

$$|\widetilde{a}(\nu)|^2 = 4\pi\sigma_t^2 I_0 e^{-8\pi^2\sigma_t^2\nu^2}. \tag{31}$$

Setting the spectral intensity to half its peak gives $|\nu| = \sqrt{\ln 2}/(2\sqrt{2}\pi\sigma_t)$, hence $\Delta\nu = \sqrt{\ln 2}/(\sqrt{2}\pi\sigma_t)$ and

$$\Delta t_{\text{pulse}} \Delta\nu = 2\sqrt{2\ln 2}\sigma_t \frac{\sqrt{\ln 2}}{\sqrt{2}\pi\sigma_t} = \frac{2\ln 2}{\pi} \simeq 0.441. \tag{32}$$

From $\nu = c/\lambda$, differentiation gives $d\nu/d\lambda = -c/\lambda^2$; for a narrow spectrum centered at λ_0, $\Delta\nu \simeq (c/\lambda_0^2)\Delta\lambda$.

The optical pulse duration can be measured with a second-order autocorrelator based on second-harmonic generation (SHG). The measured intensity is a function of the relative delay between two pulse replicas. For a Gaussian pulse, the intensity autocorrelation FWHM is $\sqrt{2}$ times the pulse-intensity FWHM. The transform-limited Gaussian time-bandwidth product is $\Delta t \Delta\nu \simeq 0.441$; a larger product can indicate spectral phase/chirp, or measurement and calibration errors. A spectrometer measures wavelength, with $\Delta\nu \simeq c\Delta\lambda/\lambda_0^2$ for a narrow spectrum. This conversion and the time-bandwidth comparison assume that the fitted spectral profile is representative of the pulse spectrum and approximately Gaussian.

2 Setup

The tutorial describes a Kerr-lens mode-locked Ti:sapphire femtosecond laser. Its intensity-dependent refractive index produces self-focusing that favors pulsed operation; a prism compressor compensates dispersion and reduces pulse chirp. For autocorrelation, a Michelson interferometer creates two pulse replicas with adjustable delay and a BBO crystal generates a second-harmonic signal proportional to their temporal overlap. For the sample measurement, the beam is split into pump and probe paths; two pump beams cross on the GaAs sample and make the excitation grating. Two probe beams arranged in BOXCAR geometry provide a diffracted signal and a local oscillator. Their interference gives a heterodyne term proportional to the diffracted field, improving sensitivity over direct intensity detection. The tutorial

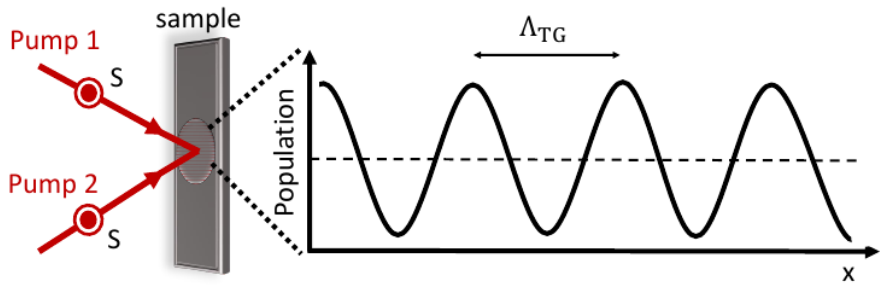


Figure 1: Two crossed pump beams create a sinusoidal excitation pattern in the sample. The grating period Λ_{TG} sets the spatial wave number $q = 2\pi/\Lambda_{TG}$ used in the diffusion analysis. Reproduced from Fig. 4.5 of the course tutorial.

describes phase modulation of the local oscillator with a rotating coverslip and pump modulation with a chopper; a lock-in amplifier extracts the component at the chopper frequency. The overall optical layout is shown in Fig. 2; the individual Michelson autocorrelator and BOXCAR detection principle are shown in Figs. 3 and 4. The provided measurement instructions clarify the period and delay calibrations used here: delay is in ps and the actual sample grating period is one quarter of the phase-mask period in the filename. Incidence angles, sample orientation, pump fluence, and sample temperature were not supplied. The data package also contains no sample thickness, pump spot size or fluence, probe fluence, detector/lock-in settings, or acquisition log; these cannot be reconstructed from the traces and should be taken from the laboratory notebook before treating the measurements as fully reproducible.

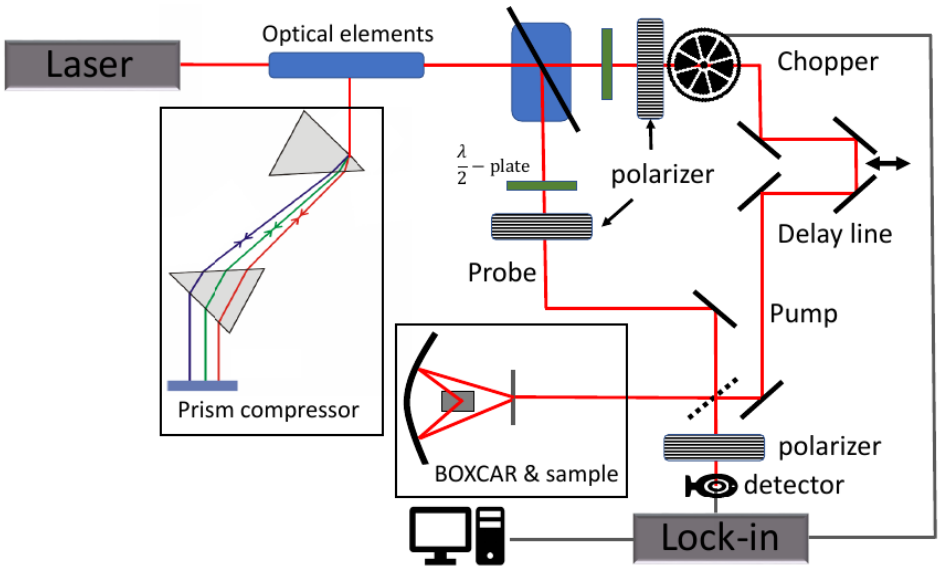


Figure 2: Tutorial schematic of the transient-reflectivity and transient-grating apparatus, including prism compressor, pump and probe paths, delay line, chopper, BOXCAR sample geometry, detector, and lock-in amplifier. Reproduced from Fig. 4.7 of the course tutorial.

3 Measurement and analysis

The analysis script `analysis/analyze.py` screens complete TG scans, averages the scans retained, estimates pointwise standard errors, fits the averaged TG and pump–probe signals

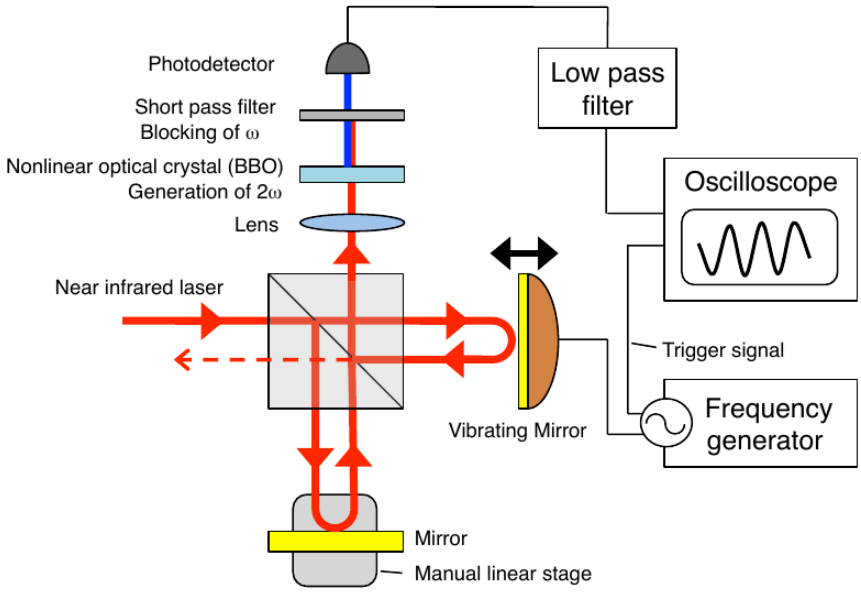


Figure 3: Michelson autocorrelator schematic: the vibrating mirror varies the relative delay between pulse replicas, and a BBO crystal generates the detected second harmonic. Reproduced from Fig. 4.1 of the course tutorial.

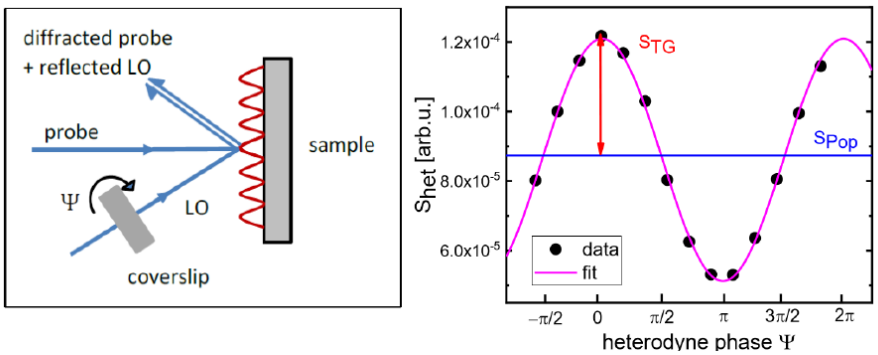


Figure 4: BOXCAR heterodyne detection of the diffracted probe and reflected local oscillator, with the measured signal illustrated as a function of heterodyne phase. Reproduced from Fig. 4.6 of the course tutorial.

to a biexponential plus offset, fits the spectrum to a Gaussian plus constant background, and evaluates the autocorrelation candidates. A 300-replicate nonparametric bootstrap (fixed random seed 20261019) resamples retained complete scans within each file, recomputes the mean traces and biexponential fits, and propagates the slow rates through the finite-q regression. Its percentile intervals quantify repeat-scan variability conditional on the retained scans and chosen fit model; they do not include calibration or model uncertainty. For each scan j, a quality score is the root-mean-square deviation from the row-wise median trace, divided pointwise by $\max(1.4826\,\mathrm{MAD}_t, s_{\mathrm{floor}})$, where s_{floor} is the lower quartile of the robust row scales over delays $t \geq 0.5\,\mathrm{ps}$. The scan scores are compared using the modified score $z_j = 0.6745(s_j - \mathrm{median}\, s)/\,\mathrm{MAD}(s)$; scans with $z_j > 3.5$ are excluded. This flags scans 4 and 10 in the phase-mask-period-8 file (modified scores 3.51 and 5.26; 2 of 44 screened TG scans overall); scan 4 is just above the cutoff, while scan 10 is a stronger outlier. The remaining scans are averaged. The five pump–probe scans show no strongly divergent scan and are kept; their small number does not support a robust scan-level cutoff. Figure 8 shows the scan-level decision, and Fig. 9 directly compares the two flagged traces with the retained-scan distribution and pointwise median. Scan 10 has a pronounced negative excursion near time zero and an elevated decay tail; scan 4 remains above the median through much of the early decay before crossing below it at later delays. This visual comparison supports the statistical flags, while not identifying their physical cause. The supplied files contain no acquisition log or instrument-status record identifying a physical cause for scans 4 or 10. Their exclusion is statistical: scan 10 is a strong whole-trace deviation, while scan 4 is marginally above the preset threshold. We do not attribute either deviation to a specific alignment, detector, or laser fault. The instructions identify the delay axis as ps and the actual sample period as one quarter of the phase-mask suffix, so $\Lambda_{\mathrm{sample}} = \Lambda_{\mathrm{mask}}/4$ and $q = 2\pi/\Lambda_{\mathrm{sample}}$. Biexponential fits use delays $t \geq 0.5\,\mathrm{ps}$ and the form $S(t) = C + A_f e^{-t/\tau_f} + A_s e^{-t/\tau_s}$ with positive amplitudes and ordered lifetimes $\tau_f < \tau_s$. Pointwise standard errors are shown for the retained-scan means. For the autocorrelation traces, `vor2` is not a full-window candidate because its scan truncates the envelope; the complete candidates are compared by normalized fit residual, and the lowest-residual trace from each before/after group is used for the reported duration.

4 Part I: Pulse duration and spectral bandwidth

The measured second-order autocorrelation is given in the tutorial by Eq. (4.2). With $I(t) = |E(t)|^2$, its full expression is

$$\begin{aligned} I_{\mathrm{AC}}(\tau) = & \int_{-\infty}^{\infty} \left[I^2(t) + I^2(t-\tau)\right] \mathrm{d}t \\ & + \int_{-\infty}^{\infty} \left[I(t) + I(t-\tau)\right] \mathrm{Re}\{E(t)E^*(t-\tau)\} \,\mathrm{d}t \\ & + \int_{-\infty}^{\infty} \mathrm{Re}\left\{E^2(t)E^{*2}(t-\tau)\right\} \,\mathrm{d}t \\ & + \int_{-\infty}^{\infty} I(t)I(t-\tau)\,\mathrm{d}t. \end{aligned} \tag{4.2}$$

For explicit evaluation, take a Gaussian field

$$E(t) = E_0 e^{-t^2/(4\sigma^2)} e^{-i\omega_0 t}, \qquad I(t) = |E(t)|^2 = I_0 e^{-t^2/(2\sigma^2)}, \qquad I_0 = |E_0|^2. \tag{33}$$

The Gaussian integral used below is obtained by squaring it and converting to polar coordinates:

$$J = \int_{-\infty}^{\infty} e^{-u^2}\, du, \tag{34}$$

$$J^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(u^2+v^2)}\, du\, dv = \int_0^{2\pi} d\varphi \int_0^{\infty} e^{-r^2} r\, dr = 2\pi \left[-\frac{e^{-r^2}}{2}\right]_0^{\infty} = \pi, \tag{35}$$

$$J = \sqrt{\pi}. \tag{36}$$

The four integrals in Eq. (4.2) then evaluate as follows. The first, delay-independent term is

$$T_0 = \int_{-\infty}^{\infty} [I^2(t) + I^2(t-\tau)]\, dt \tag{37}$$

$$= 2I_0^2 \int_{-\infty}^{\infty} e^{-t^2/\sigma^2}\, dt = 2I_0^2\sqrt{\pi}\sigma. \tag{38}$$

For the fundamental-frequency term, first multiply each intensity by the field overlap. The carrier factor is

$$E(t)E^*(t-\tau) = I_0 e^{-[t^2+(t-\tau)^2]/(4\sigma^2)} e^{-i\omega_0\tau}. \tag{39}$$

The two overlap integrals are equal after completing the square:

$$\int_{-\infty}^{\infty} I(t)E(t)E^*(t-\tau)\, dt = I_0^2 e^{-i\omega_0\tau} \int e^{-[3t^2+(t-\tau)^2]/(4\sigma^2)}\, dt \tag{40}$$

$$= I_0^2 e^{-i\omega_0\tau} e^{-3\tau^2/(16\sigma^2)} \int e^{-(t-\tau/4)^2/\sigma^2}\, dt \tag{41}$$

$$= I_0^2\sqrt{\pi}\sigma e^{-3\tau^2/(16\sigma^2)} e^{-i\omega_0\tau}, \tag{42}$$

$$\int_{-\infty}^{\infty} I(t-\tau)E(t)E^*(t-\tau)\, dt = I_0^2\sqrt{\pi}\sigma e^{-3\tau^2/(16\sigma^2)} e^{-i\omega_0\tau}. \tag{43}$$

Taking the real part of their sum gives

$$T_1 = 2I_0^2\sqrt{\pi}\sigma e^{-3\tau^2/(16\sigma^2)} \cos(\omega_0\tau). \tag{44}$$

For the second-harmonic field-overlap term,

$$E^2(t)E^{*2}(t-\tau) = I_0^2 e^{-[t^2+(t-\tau)^2]/(2\sigma^2)} e^{-i2\omega_0\tau}, \tag{45}$$

$$T_2 = \mathrm{Re} \int_{-\infty}^{\infty} E^2(t)E^{*2}(t-\tau)\, dt \tag{46}$$

$$= I_0^2 e^{-\tau^2/(4\sigma^2)} \cos(2\omega_0\tau) \int e^{-(t-\tau/2)^2/\sigma^2}\, dt \tag{47}$$

$$= I_0^2\sqrt{\pi}\sigma e^{-\tau^2/(4\sigma^2)} \cos(2\omega_0\tau). \tag{48}$$

The final intensity-overlap term has the same Gaussian integral without the cosine:

$$T_3 = \int_{-\infty}^{\infty} I(t)I(t-\tau)\, dt \tag{49}$$

$$= I_0^2 e^{-\tau^2/(4\sigma^2)} \int e^{-(t-\tau/2)^2/\sigma^2}\, dt \tag{50}$$

$$= I_0^2\sqrt{\pi}\sigma e^{-\tau^2/(4\sigma^2)}. \tag{51}$$

Combining these evaluated terms gives the Gaussian-pulse form of Eq. (4.2):

$$I_{\mathrm{AC}}(\tau) = 2I_0^2\sqrt{\pi}\sigma + 2I_0^2\sqrt{\pi}\sigma e^{-3\tau^2/(16\sigma^2)} \cos(\omega_0\tau) + I_0^2\sqrt{\pi}\sigma e^{-\tau^2/(4\sigma^2)}[1 + \cos(2\omega_0\tau)]. \tag{52}$$

Thus the integrals explicitly produce fringes at ω_0 and $2\omega_0$. For the measured trace, the fit uses a common Gaussian envelope for the slowly varying terms as a phenomenological approximation to these overlap functions; it fits

$$S(x) = S_0 + e^{-(x-x_0)^2/(2\sigma_x^2)} \left[A_0 + A_1 \cos(2\pi f(x-x_0) + \phi_1) + A_2 \cos(4\pi f(x-x_0) + \phi_2)\right]. \quad (53)$$

In a scan coordinate x with fringe frequency f, one optical cycle takes $1/f$ on the recorded axis and λ_0/c in delay. Therefore

$$\tau - \tau_0 = \frac{\lambda_0}{c} f(x - x_0), \quad (54)$$

$$\omega_0(\tau - \tau_0) = \frac{2\pi c}{\lambda_0}\frac{\lambda_0}{c} f(x - x_0) = 2\pi f(x - x_0), \quad (55)$$

$$2\omega_0(\tau - \tau_0) = 4\pi f(x - x_0), \quad (56)$$

which gives the two cosine arguments in the fit. The fitted Gaussian intensity envelope is

$$I_{\mathrm{env}}(x) = A_{\mathrm{env}} \exp\left[-\frac{(x-x_0)^2}{2\sigma_x^2}\right], \qquad A_{\mathrm{env}} = A_0 + |A_1| + |A_2|. \quad (57)$$

In the tutorial's field-envelope notation, its square root is

$$E_{\mathrm{env}}(x) = \sqrt{I_{\mathrm{env}}(x)} = \sqrt{A_{\mathrm{env}}} \exp\left[-\frac{(x-x_0)^2}{4\sigma_x^2}\right], \qquad E_{\mathrm{env}}^2(x) = I_{\mathrm{env}}(x). \quad (58)$$

The appendix plot overlays the normalized $E_{\mathrm{env}}(x)$ (the square root of the fitted intensity-envelope) on the raw signal. The fit envelope has standard deviation σ_x in recorded coordinate. The local scan factor is $d\tau/dx = f\lambda_0/c$, so $\sigma_\tau = (f\lambda_0/c)\sigma_x$. Its FWHM and the pulse FWHM are

$$\Delta t_{\mathrm{AC}} = 2\sqrt{2\ln 2}\,\sigma_\tau = 2\sqrt{2\ln 2}\,\sigma_x f\lambda_0/c, \quad (59)$$

$$\Delta t_{\mathrm{pulse}} = \frac{\Delta t_{\mathrm{AC}}}{\sqrt{2}}. \quad (60)$$

This conversion assumes a locally constant delay-scan speed. The fitted traces are shown in Fig. 12. We exclude `vor2`: its recorded scan ends at 2.8 ms, compared with 14 ms for `vor1` and `vor3`, leaving the outer autocorrelation wings incompletely recorded (Fig. 11). Its sampling interval also differs by a factor of five, so its scan calibration is not directly comparable. Including it would bias the before-compressor pulse-width estimate.

The measured spectrum and Gaussian fit are shown in Fig. 5. The spectrum file spans approximately 345 nm to 1041 nm. We fitted all recorded points with

$$S(\lambda) = B + A \exp\left[-\frac{(\lambda-\lambda_0)^2}{2\sigma_\lambda^2}\right]. \quad (61)$$

The peak above baseline is A and the fitted peak level is $B + A$. At half peak height above baseline, $S - B = A/2$, so

$$e^{-(\lambda-\lambda_0)^2/(2\sigma_\lambda^2)} = \frac{1}{2}, \quad (62)$$

$$|\lambda - \lambda_0| = \sqrt{2\ln 2}\,|\sigma_\lambda|, \quad (63)$$

$$\Delta\lambda_{\mathrm{FWHM}} = 2\sqrt{2\ln 2}\,|\sigma_\lambda|. \quad (64)$$

The fit gives a centre of 788.66 nm, peak intensity of 13 302 a u , median measured intensity of 2501 a u , and fitted half-maximum level of 7902 a u ; $\Delta\lambda_{\mathrm{FWHM}} \approx 10.92$ nm. Applying $\Delta\nu \simeq$

$c\Delta\lambda/\lambda_0^2$ gives $\Delta\nu \approx 5.26\,\mathrm{THz}$. The Gaussian time-bandwidth result derived above gives $\Delta t_{\mathrm{TL}} = 0.441/\Delta\nu \approx 83.8\,\mathrm{fs}$. The median is calculated from all measured spectrum points; it is plotted as a descriptive reference and is not the fitted background B.

The normalized fit residuals for the complete autocorrelation candidates are 0.1222 (`vor1`), 0.1214 (`vor3`), 0.0922 (`nach1`), 0.0914 (`nach2`), and 0.0909 (`nach3`). Thus `vor3` and `nach3` are selected as the cleanest traces under this criterion. Their fitted pulse durations are 81.7 fs and 79.0 fs, respectively. These are individual selected-trace estimates, not averages over repeats; the corresponding time-bandwidth products are about 0.43 and 0.42. Systematic uncertainty from scan nonlinearity, model assumptions, and instrument response remains unquantified. The reported spectral bandwidth uncertainty ($\pm 0.03\,\mathrm{nm}$) and autocorrelation fit uncertainties ($\pm 0.50\,\mathrm{fs}$ before and $\pm 0.36\,\mathrm{fs}$ after) are formal nonlinear-fit covariance errors only; they exclude calibration, model, and candidate-selection uncertainty. The inferred durations are slightly below the Gaussian transform limit, so these small formal errors should not be read as total experimental uncertainties.

5 Part II: Hot-carrier cooling and electron–phonon coupling

The tutorial asks for a hot-carrier cooling time and an average electron–phonon coupling coefficient. These data do not support either quantity as a physical measurement: the reflectivity traces record an optical response, not the carrier temperature or energy distribution, and the biexponential fit is phenomenological. In addition, the fitted interval starts at 0.5 ps, so it cannot resolve the subpicosecond thermalization and scattering dynamics. We therefore do not identify τ_f with a cooling constant or calculate an electron–phonon coupling coefficient from it. For context, GaAs thermalization has been reported on a roughly 200 fs scale, and average electron–LO-phonon scattering around 165 fs for injected densities below $10^{17}\,\mathrm{cm}^{-3}$; these are microscopic times and are not directly comparable to our fitted reflectivity lifetimes [2, 4]. A defensible cooling constant and coupling estimate would require a calibrated carrier-temperature or energy-versus-delay measurement and the relevant excitation conditions.

6 Part III: Transient-reflectivity and transient-grating decay times

The retained-scan means and biexponential fits are shown in Fig. 6; Fig. 7 shows residuals normalized by the fitted-window signal range. The normalized residual RMS is 1.26% for pump–probe and 0.71–0.81% for TG. The scan-level screen and direct comparison of excluded traces appear in Figs. 8 and 9. The residuals show that the biexponential describes the averaged traces over the fitted interval, but do not establish a unique microscopic assignment.

Table 1 gives the fitted fast and slow decay times. The data columns are averaged only after the scan screen; the table gives 95% bootstrap intervals from resampling retained scans, conditional on the selected traces and model. The sample periods use the supplied quarter-period correction, and the first column of each transient file is interpreted in ps.

Table 1: Biexponential fit lifetimes and 95% repeat-scan bootstrap intervals. The filename value is the phase-mask period; the sample period is one quarter of that value.

File	$\Lambda_{\mathrm{sample}}$ (µm)	q (m^{-1})	τ_f (ps; 95% CI)	τ_s (ps; 95% CI)
`GaAs_RT_PP_X.dat`	–	0	0.717 [0.698, 0.743]	7.878 [7.753, 8.040]
`GaAs_RT_5.6_X.dat`	1.400	4.488×10^6	1.206 [1.031, 1.445]	7.205 [6.412, 8.771]
`GaAs_RT_6.5_X.dat`	1.625	3.867×10^6	1.298 [1.157, 1.503]	9.200 [8.156, 10.960]
`GaAs_RT_8_X.dat`	2.000	3.142×10^6	1.363 [1.064, 1.983]	9.496 [7.933, 12.250]
`GaAs_RT_12_X.dat`	3.000	2.094×10^6	2.060 [1.708, 2.463]	13.283 [11.970, 15.170]

7 Part IV: Recombination, ambipolar diffusivity, and mobility

For the finite-q TG measurements, the slow-component rates are regressed against q^2 as motivated by the diffusion equation. The slope is $3.71 \times 10^{-15}\,\mathrm{m^2\,ps^{-1}}$ ($R^2 = 0.934$), giving a TG-only ambipolar diffusion estimate $D_a = (3.71 \pm 0.70) \times 10^{-3}\,\mathrm{m^2\,s^{-1}}$, where the stated error is the standard error of the four-point regression slope. Resampling complete retained scans gives a 95% bootstrap interval $[2.31, 4.96] \times 10^{-3}\,\mathrm{m^2\,s^{-1}}$. This interval includes repeat-scan variation but not calibration or model uncertainty. Retaining borderline scan 4 while still excluding scan 10 changes the estimate by about 2% to $3.78 \times 10^{-3}\,\mathrm{m^2\,s^{-1}}$ (Fig. 10).

The finite-q intercept is $0.0612\,\mathrm{ps^{-1}}$, which would correspond to a nominal recombination time of 16.3 ps under the single-intercept model. However, the pump–probe slow rate is $1/(7.878\,\mathrm{ps}) = 0.127\,\mathrm{ps^{-1}}$, about twice the TG extrapolation; including $q = 0$ in the regression reduces R^2 to 0.161. The pump–probe and TG signals can weight multiple relaxation channels differently, so this mismatch cautions against interpreting the extrapolated intercept as a measured recombination lifetime. The biexponential amplitudes and lifetimes are phenomenological, and the four-period fit supports only an exploratory TG diffusion estimate.

Using the Einstein relation $\mu_a = eD_a/(k_BT)$ and assuming room temperature $T = 300\,\mathrm{K}$ gives a conditional ambipolar mobility estimate of about $1.44 \times 10^3\,\mathrm{cm^2\,V^{-1}\,s^{-1}}$; converting the regression standard error gives approximately $\pm 0.27 \times 10^3\,\mathrm{cm^2\,V^{-1}\,s^{-1}}$. The repeat-scan bootstrap interval corresponds to roughly 0.89–$1.92 \times 10^3\,\mathrm{cm^2\,V^{-1}\,s^{-1}}$. This conversion assumes non-degenerate carriers at a common temperature and that the RT filename denotes 300 K; the actual sample temperature was not logged. A bulk-GaAs pump–probe study reports ambipolar diffusivity near $20\,\mathrm{cm^2/s}$ at room temperature, compared with our $37\,\mathrm{cm^2/s}$ estimate. The values are the same order of magnitude, but ours is about 1.9 times larger; the bootstrap range and missing carrier-density and calibration information prevent a strong discrepancy claim [3]. Density-dependent many-body effects can also alter GaAs diffusivity [6].

The dominant unresolved uncertainties are grating-period and delay calibration, the biexponential model and fit interval, and the unrecorded sample temperature, pump fluence, carrier density, spot size, sample thickness, and orientation. These details are needed for a like-for-like literature comparison and full reproducibility; the supplied data files do not contain them.

8 Conclusion

The four tutorial analysis tasks lead to different levels of confidence. The pulse spectrum and autocorrelation give durations near 80 fs and a transform limit of 83.8 fs, although the fitted durations falling slightly below that limit indicate unquantified systematic uncertainty. The supplied reflectivity data do not independently determine hot-carrier cooling or an electron–phonon coupling coefficient. Biexponential fits provide effective transient-reflectivity and TG decay times, but do not uniquely identify microscopic channels. The finite-q TG data give an exploratory $D_a = (3.71 \pm 0.70) \times 10^{-3}\,\mathrm{m^2\,s^{-1}}$ estimate and, assuming 300 K, a conditional mobility near $1.44 \times 10^3\,\mathrm{cm^2\,V^{-1}\,s^{-1}}$. The inconsistent pump–probe intercept, limited number of grating periods, and unavailable run metadata limit these transport results.

References

[1] Institute of Physics II, University of Cologne, *M2.10: Ultrafast Transient Grating Spectroscopy*, practical-course tutorial, version December 2019, especially Eqs. (4.2)–(4.14) and Secs. 5–6.

[2] S. Hunsche, H. Heesel, A. Ewertz, H. Kurz, and J. H. Collet, "Spectral-hole burning and carrier thermalization in GaAs at room temperature, *Physical Review B* **48**, 17818 (1993), doi:10.1103/PhysRevB.48.17818.

[3] B. A. Ruzicka, L. K. Werake, H. Samassekou, and H. Zhao, "Ambipolar diffusion of photo-excited carriers in bulk GaAs, *Applied Physics Letters* **97**, 262119 (2010), doi:10.1063/1.3533664.

[4] J. A. Kash, J. C. Tsang, and J. M. Hvam, "Subpicosecond Time-Resolved Raman Spectroscopy of LO Phonons in GaAs," *Physical Review Letters* **54**, 2151 (1985), doi:10.1103/PhysRevLett.54.2151.

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A Plots

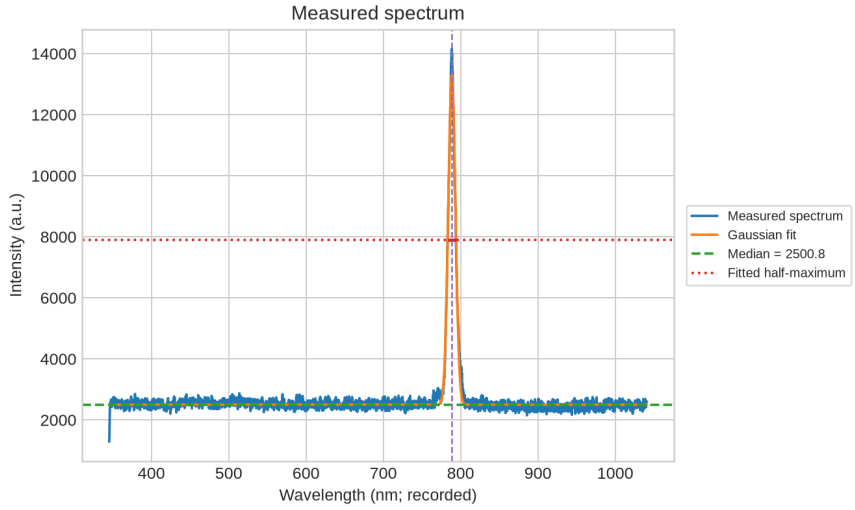


Figure 5: Measured spectrum, Gaussian fit, median level, and fitted half-maximum bandwidth.

AI Use Disclosure

Generative AI was used to assist with data-analysis code, plotting, and editing the presentation of this report. The authors reviewed the analysis, calculations, figures, and final text and remain responsible for their accuracy and conclusions.

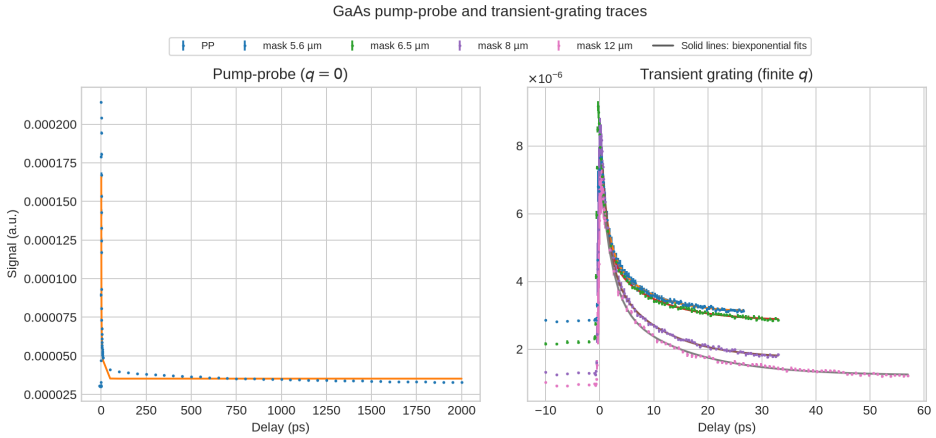


Figure 6: Mean delay traces over retained scans and biexponential fits, with delay in ps. Error bars show the standard error across retained scans.

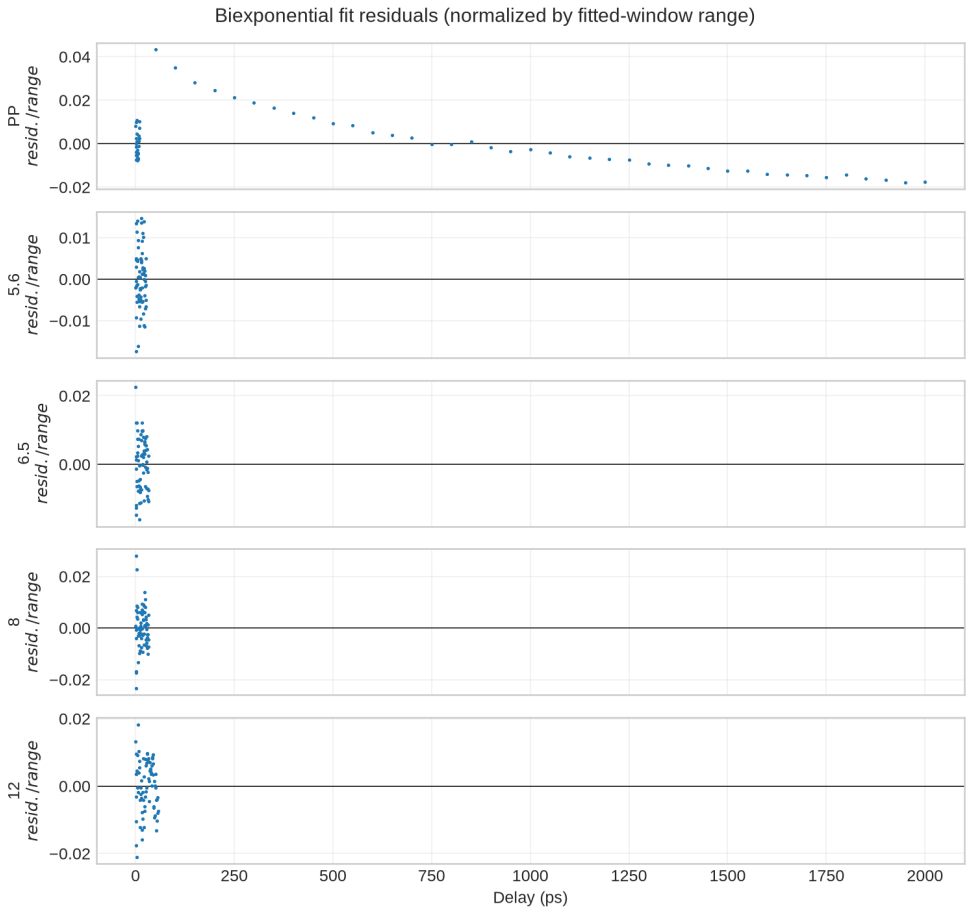


Figure 7: Biexponential fit residuals for the averaged data, normalized by the fitted-window signal range. The traces are displayed separately to make structure visible.

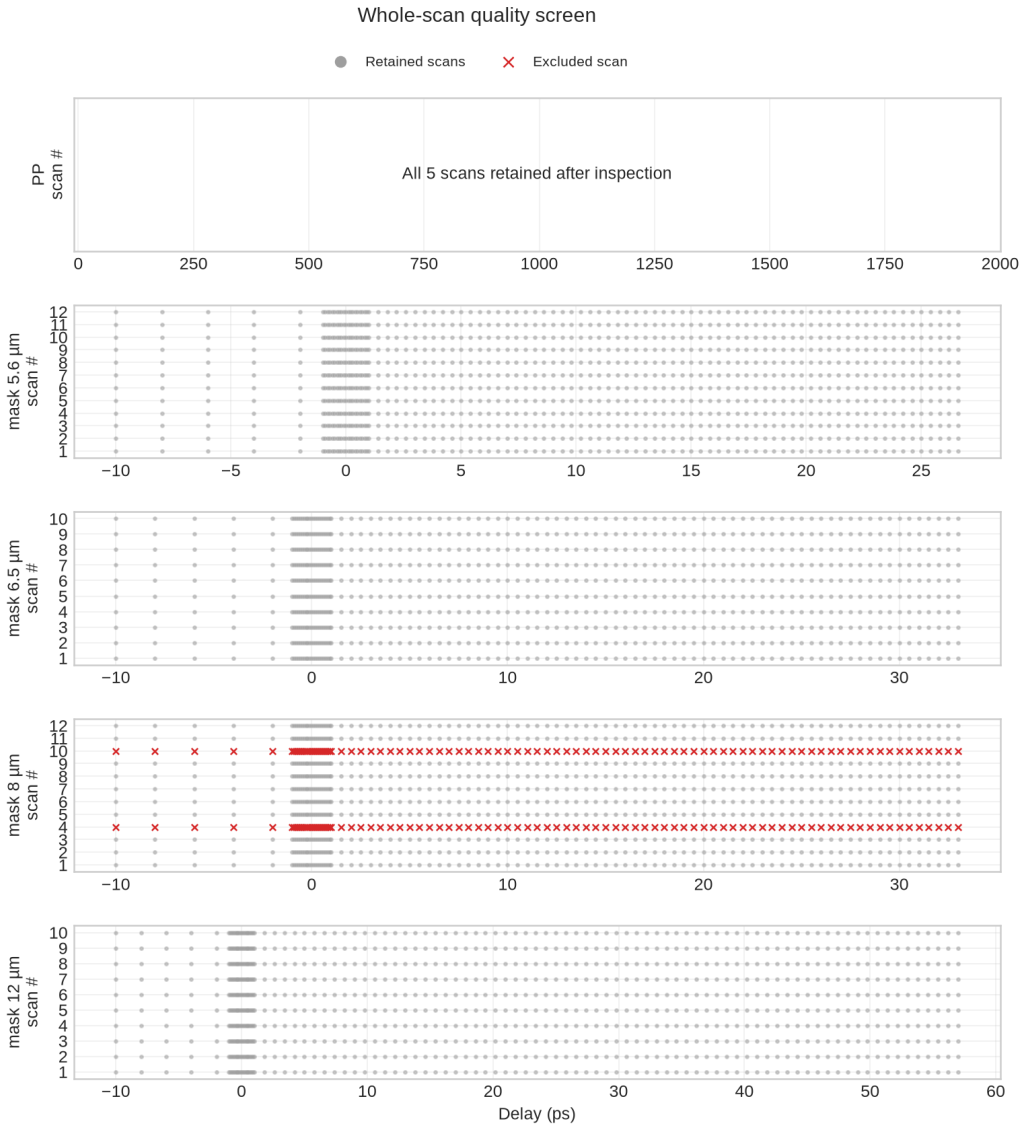


Figure 8: Whole-scan quality screen. Red crosses mark scans excluded by the robust scan-deviation criterion; all five pump–probe scans were retained after inspection.

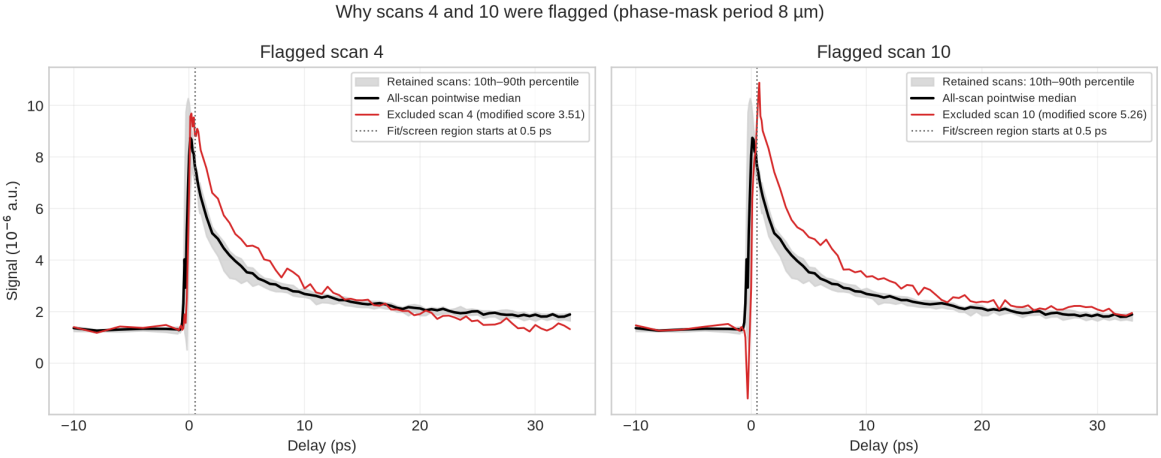


Figure 9: The two statistically flagged scans from the phase-mask-period-8 file, compared with the all-scan pointwise median and the 10th–90th percentile band of retained scans. The vertical dotted line marks the 0.5 ps fit/screen threshold. Scan 4 is borderline; scan 10 shows the larger deviation.

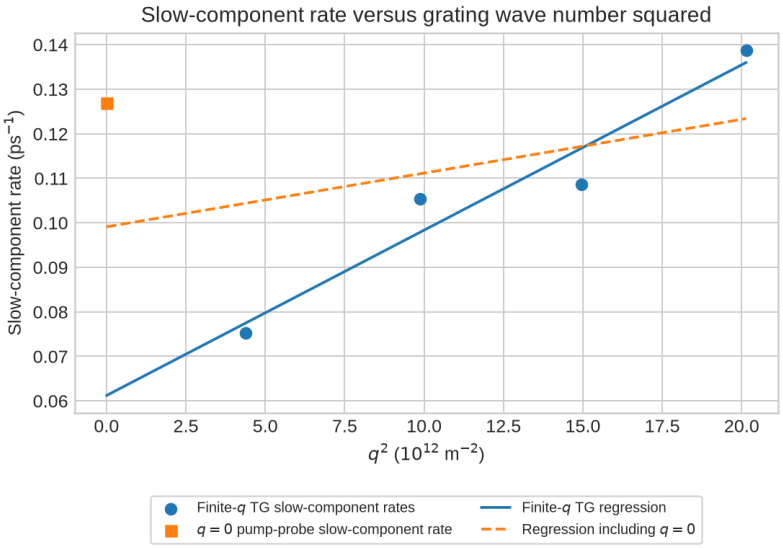


Figure 10: Slow-component rates against q^2 computed from the sample grating periods (one quarter of the phase-mask values). The finite-q line gives the TG-only diffusion estimate; including the q=0 pump–probe rate gives a poor linear fit.

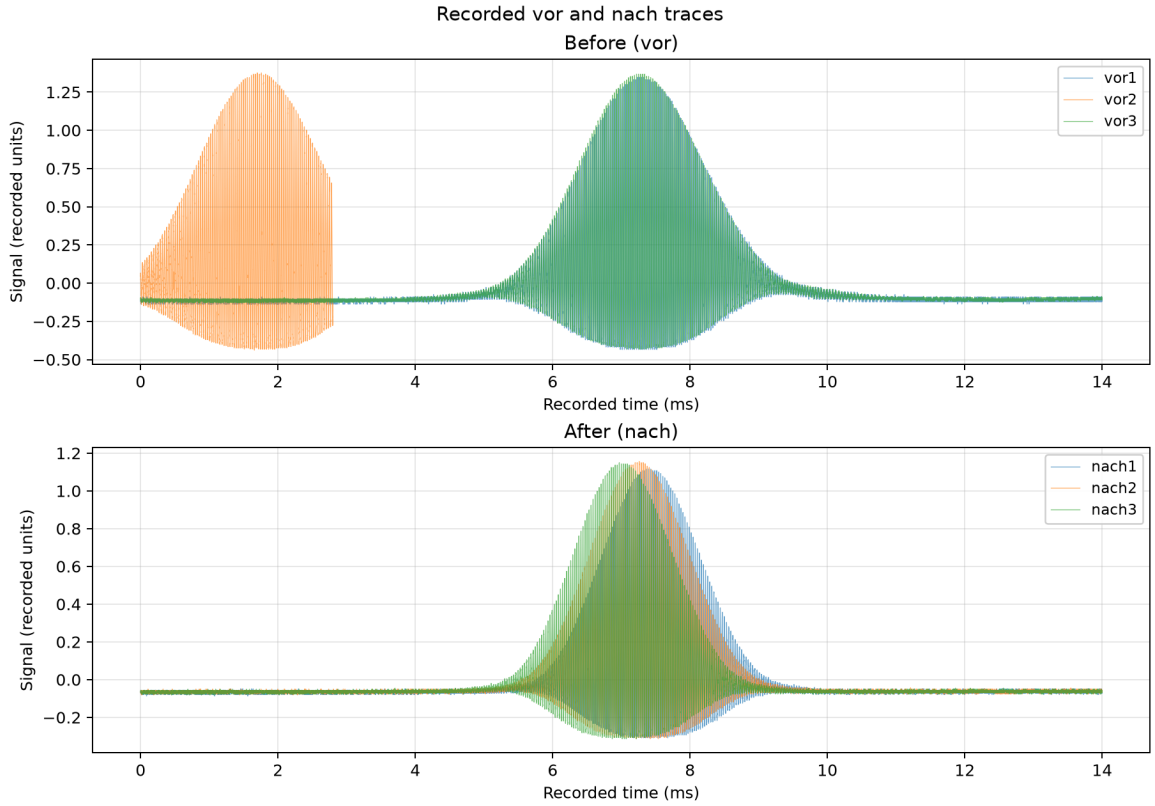


Figure 11: Recorded auxiliary traces grouped as before (vor) and after (nach). The horizontal axis uses recorded time converted to milliseconds.

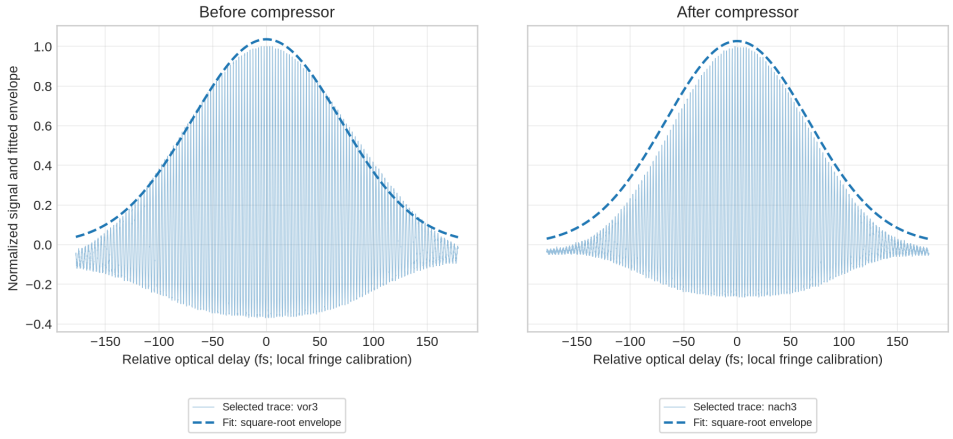


Figure 12: Selected cleanest complete before and after autocorrelation traces, `vor3` and `nach3`, with fitted square-root Gaussian envelopes. Candidate quality is ranked by normalized fit residual; the truncated `vor2` scan is excluded.